
UK Space-Based Solar Power Cost-Threshold Assessment

Report subtitle: Decision-focused techno-economic assessment for UK firm low-carbon electricity

Evidence status: Project evidence registry and official source files reviewed to 10 July 2026.

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Prepared as: A reproducible threshold assessment using the project data and model pipeline.

Disclaimer: This report identifies cost conditions for further assessment. It does not establish engineering feasibility, commercial readiness or investment merit.

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1. Executive Decision Summary

The reference case produces delivered-grid LCOE of 429 GBP/MWh. It is an analytical anchor, not a forecast or preferred design. The useful result is the threshold structure below it.

Table 1. Cost gates and their decision meaning

Cost gate	Interpretation	Decision meaning
Below 150 GBP/MWh	Broad screening ceiling	Approaches the upper comparator range; not a competitiveness claim
100-120 GBP/MWh	Firm low-carbon decision region	Warrants engineering, finance and GB system-value assessment
80 GBP/MWh or below	Conservative system-adjusted overlap	Cost is relevant, but price-basis and system-value checks still apply
60 GBP/MWh	Stringent stretch test	Requires simultaneous movement across most bottlenecks

Three conclusions drive the assessment:

- Only specific mass reaches the 150 GBP/MWh screen by itself. With all other inputs fixed, the thresholds are 1.28 kg/kW-space at 150 GBP/MWh, 0.88 at 120, and 0.62 at 100.
- Launch cost alone bottoms out at 204 GBP/MWh even at 20 GBP/kg. End-to-end efficiency alone bottoms out at 188 GBP/MWh at 35 percent efficiency.
- Reaching the 80-120 GBP/MWh region requires coordinated improvement in mass, efficiency, launch, assembly, orbital hardware and finance.

Entering 80-120 GBP/MWh would justify deeper engineering and GB power-system modelling. It would not by itself establish market competitiveness.

2. Scope and Method

The model asks what launch cost, mass intensity, hardware cost, assembly cost, efficiency, utilisation, lifetime, OPEX and financing conditions would be required for SBSP to approach UK electricity cost benchmarks.

The boundary is delivered-grid LCOE. Annualised initial CAPEX, fixed OPEX, refurbishment and variable OPEX are divided by annual delivered MWh. Delivered electricity equals capacity x 8,760 x capacity factor; required space-side power equals delivered capacity divided by end-to-end efficiency; orbital mass equals space-side power multiplied by specific mass.

The model identifies cost parity only. Engineering achievability, beam and spectrum regulation, construction delivery, commercial financeability and whole-system value require separate tests.

3. Reference Case and Cost Structure

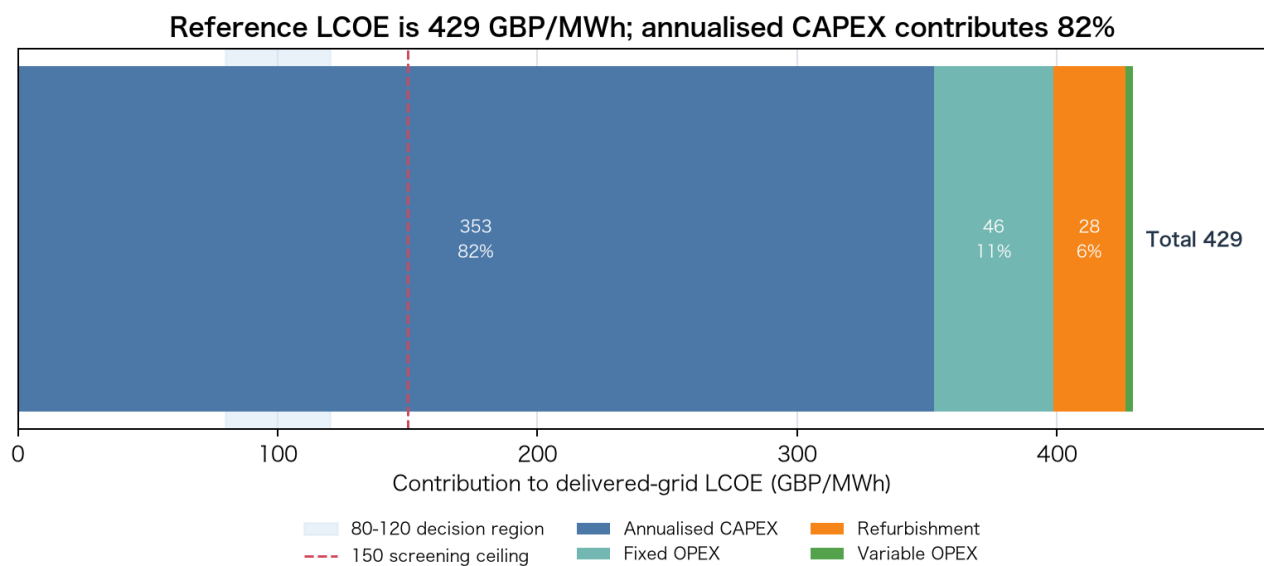
Table 2. Core reference inputs

Parameter	Reference value	Role
Delivered grid capacity	2,000 MW	Scale anchor
End-to-end efficiency	15.0%	Sets space-side power and mass
Capacity factor / availability	90.0%	Sets annual delivered electricity
Specific mass	5.0 kg/kW-space	Multiplies launch, transfer and assembly cost
Launch cost	500 GBP/kg to staging orbit	Primary transport-cost lever
Space hardware cost	0.40 GBP/W-space	Orbital manufacturing-cost lever
In-orbit assembly and deployment cost	200 GBP/kg	Mass-scaled deployment cost
WACC / discount rate	6.5%	Annualises the capital base
System lifetime	30 years	Capital recovery period

At this point, 2 GW delivered output requires 13.33 GW of space-side power and about 66,667 tonnes of orbital hardware.

Table 3. Initial CAPEX, including programme margin

Component	GBP billion
Launch	33.33
In-orbit assembly and deployment	13.33
Orbit transfer	6.67
Space segment CAPEX	5.33
Wireless power transmission	1.33
Rectenna	0.30
Grid connection	0.20
Pre-margin subtotal	60.50
Programme margin / contingency	12.10
Initial CAPEX total	72.60

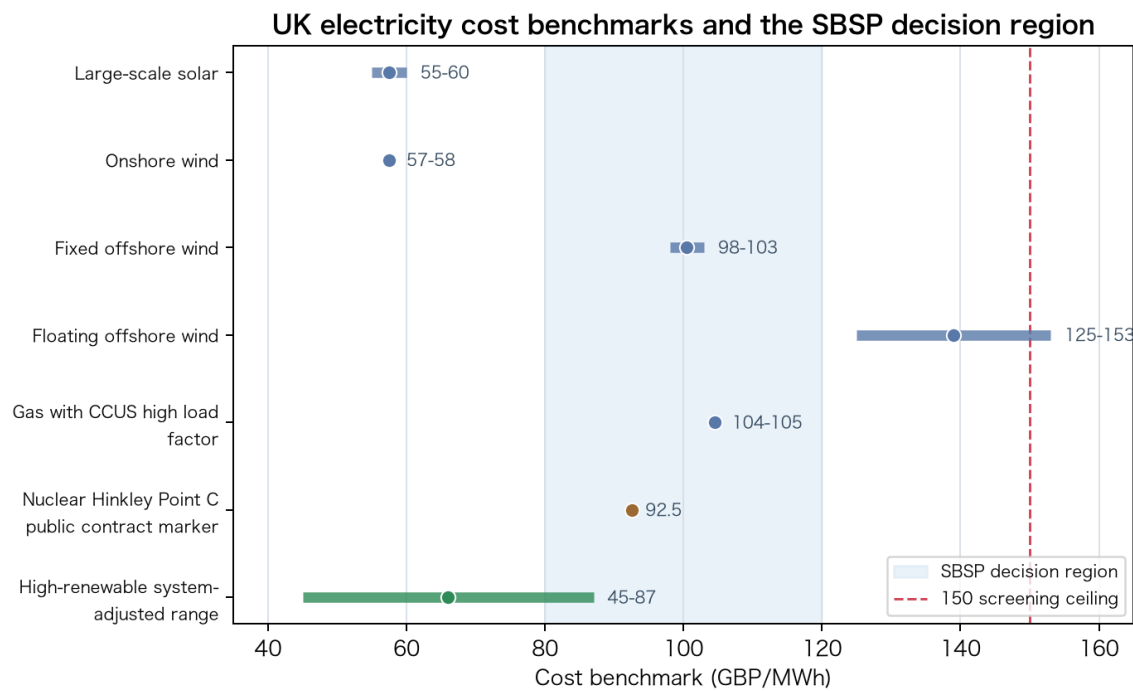
Figure 1. Reference LCOE composition

Annualised CAPEX contributes about 82 percent of reference LCOE. This is why mass, scale, hardware cost and financing dominate the threshold result.

The reference inputs are exploratory because commercial-scale SBSP has not been deployed. They answer what would have to change, not what will happen.

4. UK Cost Benchmarks

Figure 2. UK electricity cost benchmarks



DESNZ values are 2024 real GBP; BEIS system-adjusted values are 2018 real GBP; the Hinkley marker is 2012 nominal GBP.

Ranges are shown as intervals rather than zero-based bars. The 80-120 GBP/MWh region is a decision window, while 150 GBP/MWh is only a broad screening ceiling.

Table 4. Benchmark interpretation

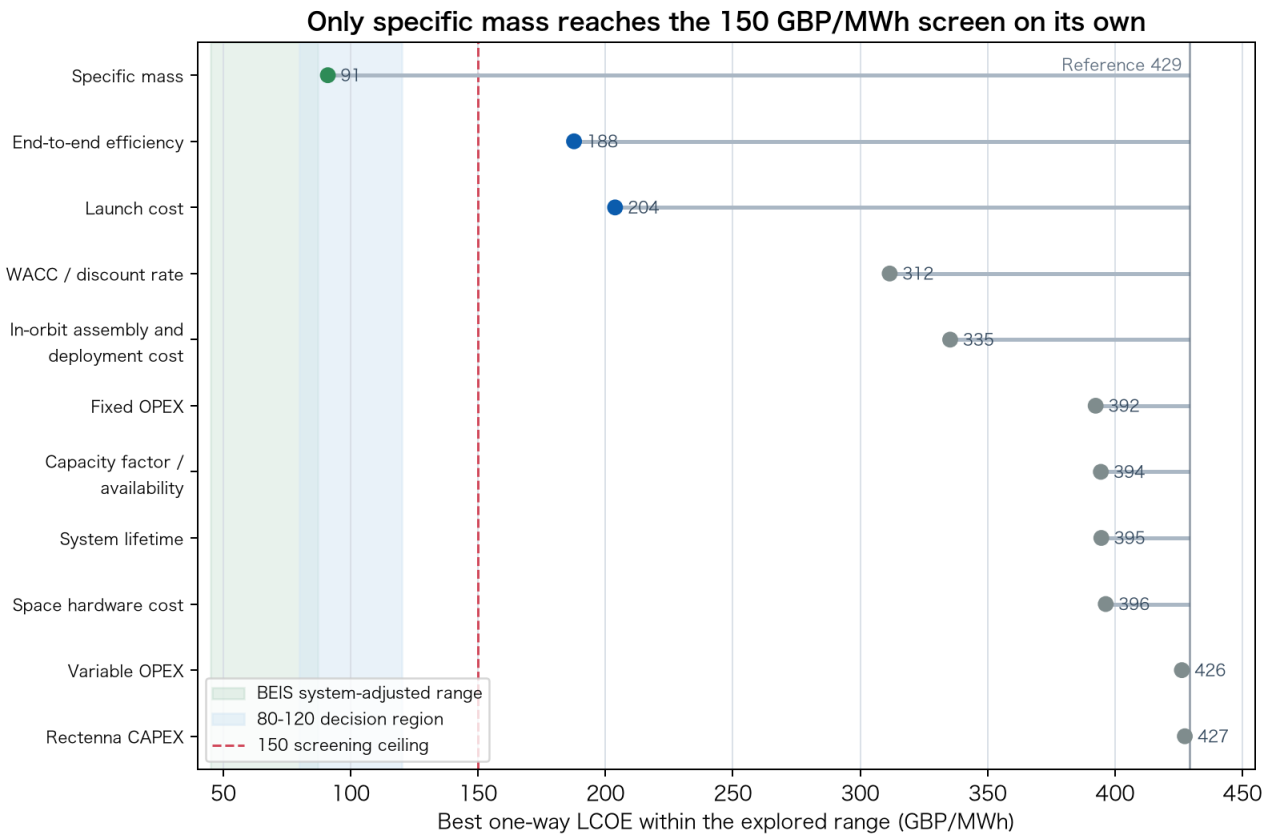
Comparator	Indicative GBP/MWh	Price basis	Use
DESNZ generation range	55-153	2024 real GBP	Generator-boundary comparison
Core mature generation range	55-113	2024 real GBP	Headline generation-cost screen
BEIS renewable system-adjusted range	45-87	2018 real GBP	Conservative indicator; not a complete GB system model
Firm low-carbon markers	92.5 contract; 104-105 CCUS	Mixed price bases	Context only; not a single like-for-like benchmark

DESNZ generation costs and BEIS enhanced-LCOE values are not on the same real-price basis. The chart labels that difference explicitly; exact overlap should not be read as price-normalised parity. Wider grid, balancing, curtailment, storage and reliability costs also require a full GB system model.

5. One-way Cost Thresholds

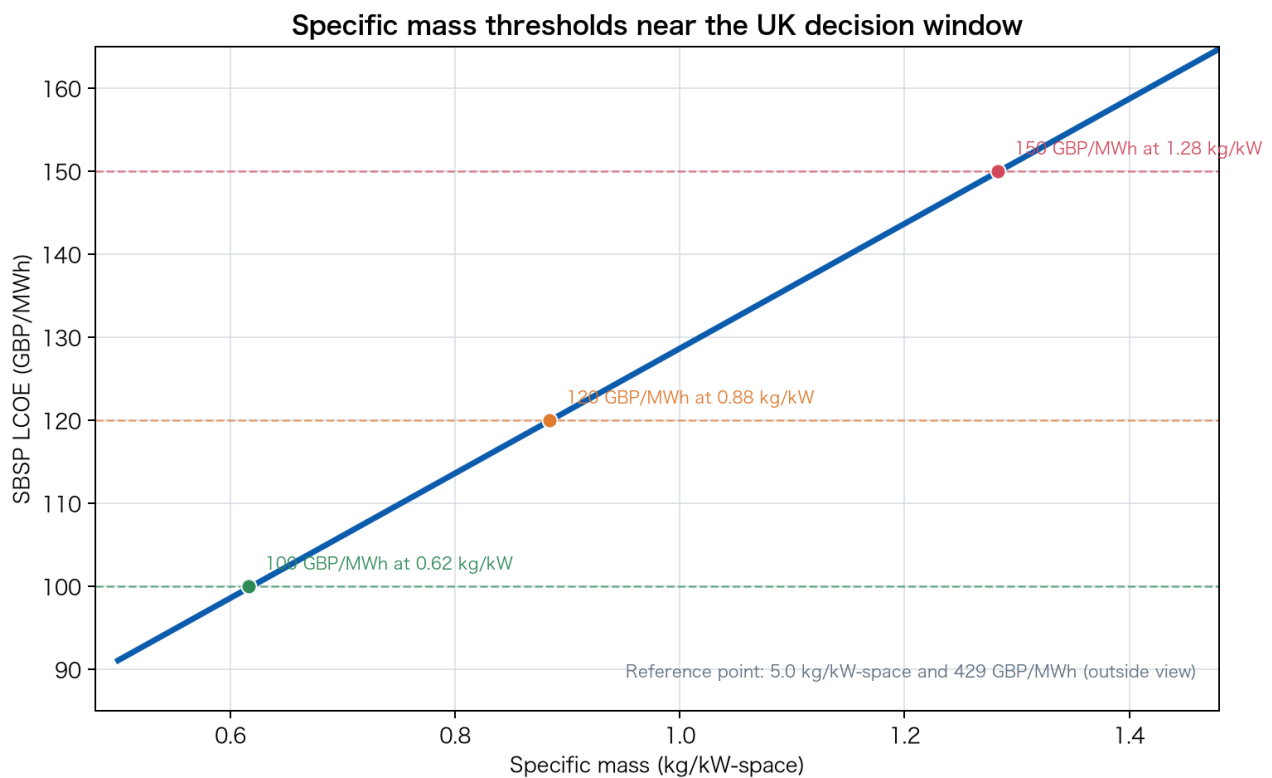
One-way analysis changes one input while holding all others at the reference point. It is a leverage test, not a development plan.

Figure 3. Best LCOE reached by each one-way change



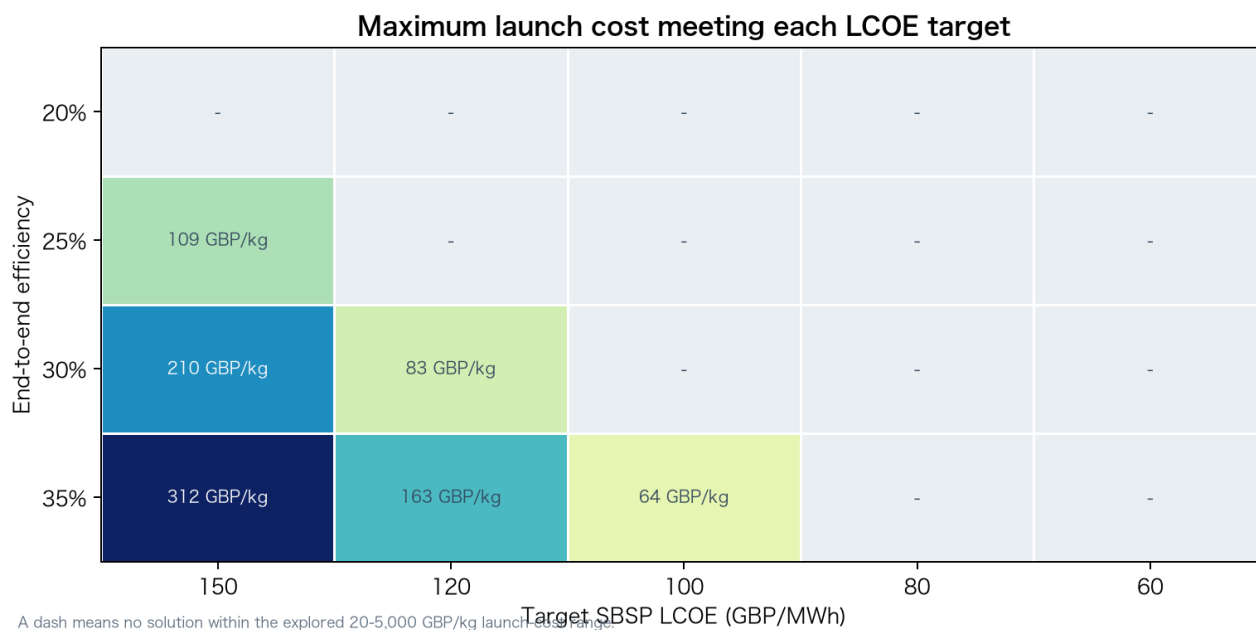
The figure removes high-cost deterioration ranges and compares only the most favourable result for each input. Specific mass is the only one-way lever that crosses the 150 GBP/MWh screen.

Figure 4. Specific-mass thresholds near the decision window



The view is restricted to the part of the curve that crosses 150, 120 and 100 GBP/MWh. The 5.0 kg/kW-space reference point is deliberately noted outside the plotting window.

Figure 5. Launch-efficiency threshold matrix



At 25 percent efficiency, 150 GBP/MWh requires launch cost of about 109 GBP/kg or lower. At 35 percent efficiency, 100 GBP/MWh requires about 64 GBP/kg. Dashes indicate no solution in the explored launch-cost range.

The pattern is unambiguous: low launch cost applied to a heavy, inefficient architecture is still too expensive. Mass and efficiency change the scale on which transport and assembly costs act.

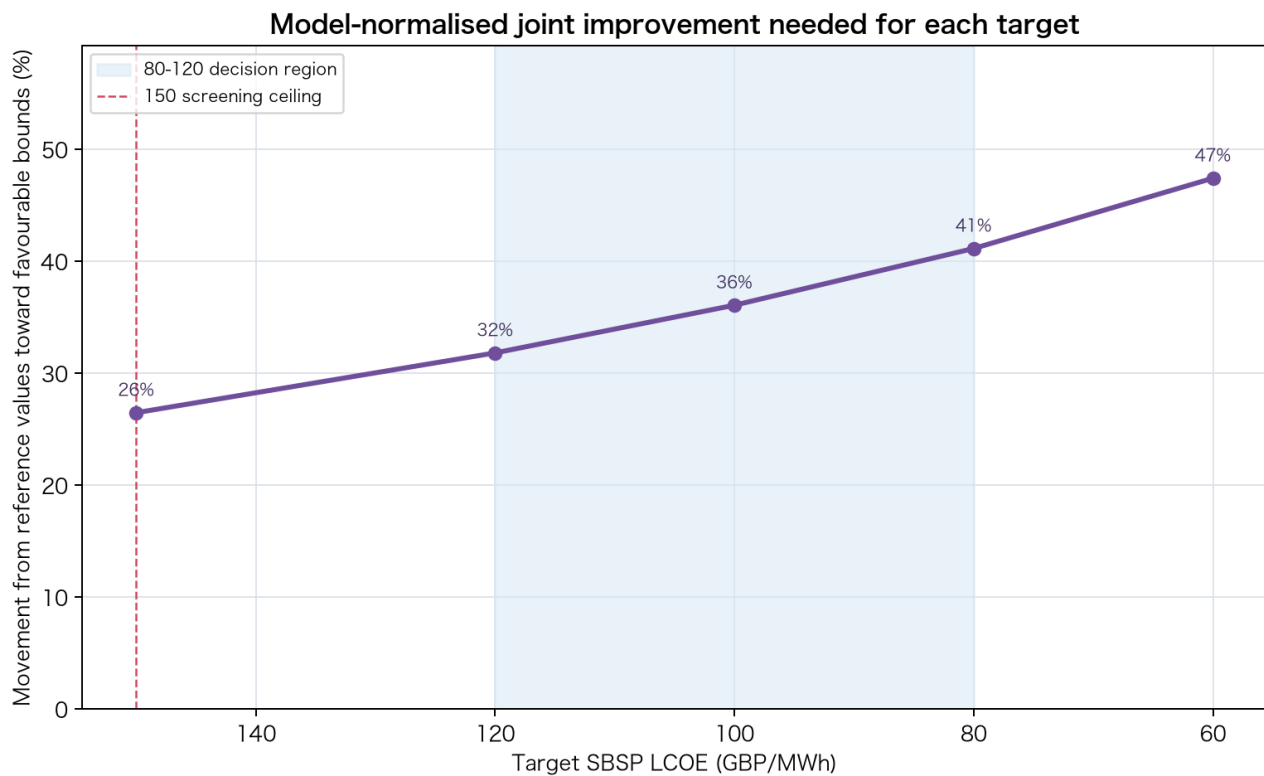
6. Combined Bottleneck Thresholds

The combined frontier moves model inputs by an equal fraction from their reference values toward the favourable ends of their documented ranges. The percentage below is a model-normalised parameter movement, not technology maturity, probability or calendar progress.

Table 5. Selected combined-threshold points

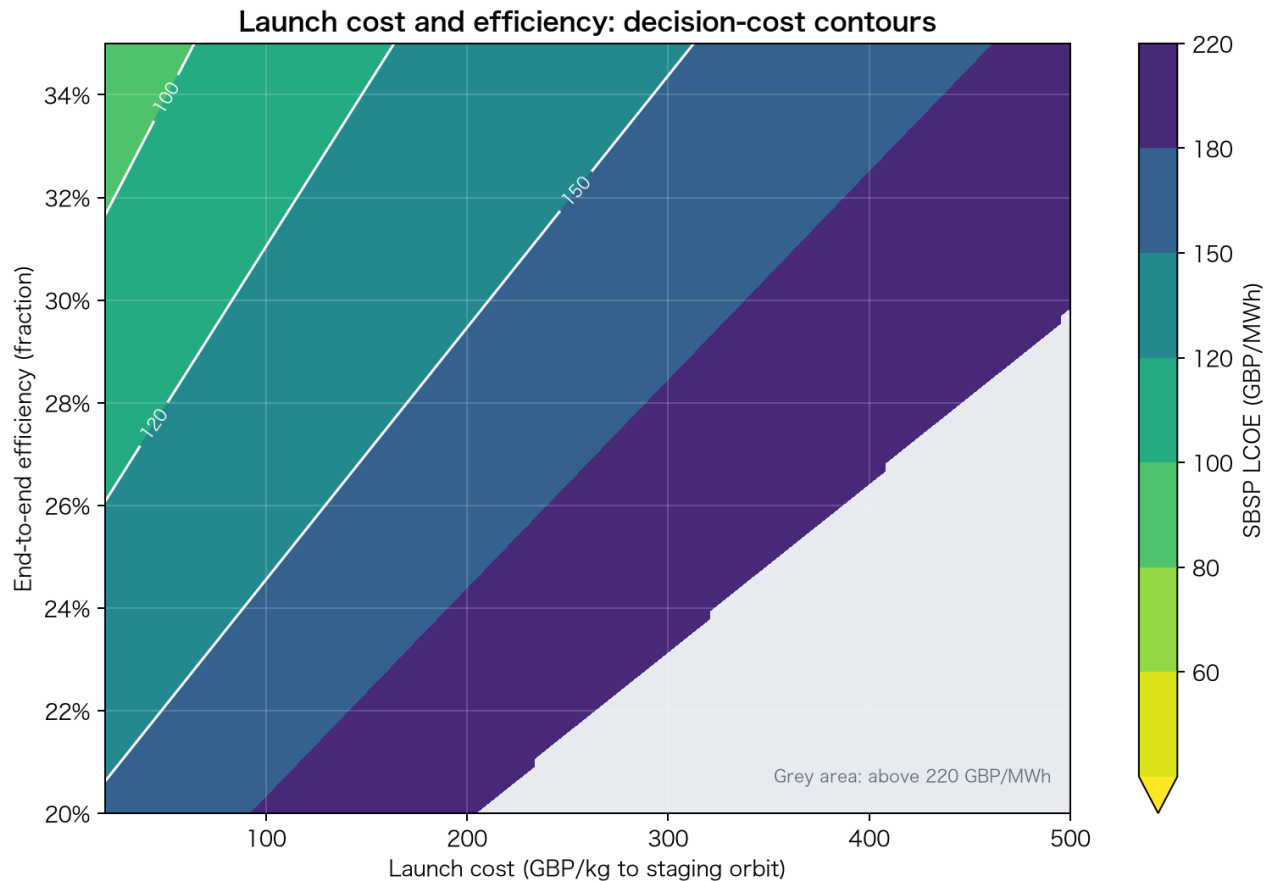
Target GBP/MWh	Normalised movement	Launch GBP/kg	Mass kg/kW	Efficiency	WACC
150	26%	373	3.81	20%	5.6%
120	32%	347	3.57	21%	5.4%
100	36%	327	3.38	22%	5.2%
80	41%	303	3.15	23%	5.1%

Figure 6. Model-normalised joint improvement



Moving from the 150 screen into the 80-120 decision region tightens several physical and financial constraints together.

Figure 7. Launch-cost and efficiency decision contours



The chart displays only 220 GBP/MWh and below; higher-cost space is neutral grey. It shows the limited launch-cost window associated with 150, 120, 100 and 80 GBP/MWh contours.

These combinations are illustrative slices through parameter space, not engineering roadmaps. Alternative slices remain available in [data/processed/alternative_combined_pathways.csv](#).

7. Decision Interpretation and Limitations

SBSP is most relevant to the UK if it can provide firm or near-firm low-carbon output. In that role, the comparison is with reliable delivered electricity, gas CCUS, nuclear-like firm supply, long-duration flexibility and the system-adjusted cost of a high-renewable system - not only bare wind and solar LCOE.

The practical decision rule is: below 150 GBP/MWh, continue threshold monitoring; in the 100-120 region, undertake integrated engineering and finance review; at 80 GBP/MWh or below, add detailed GB dispatch, network and reliability modelling.

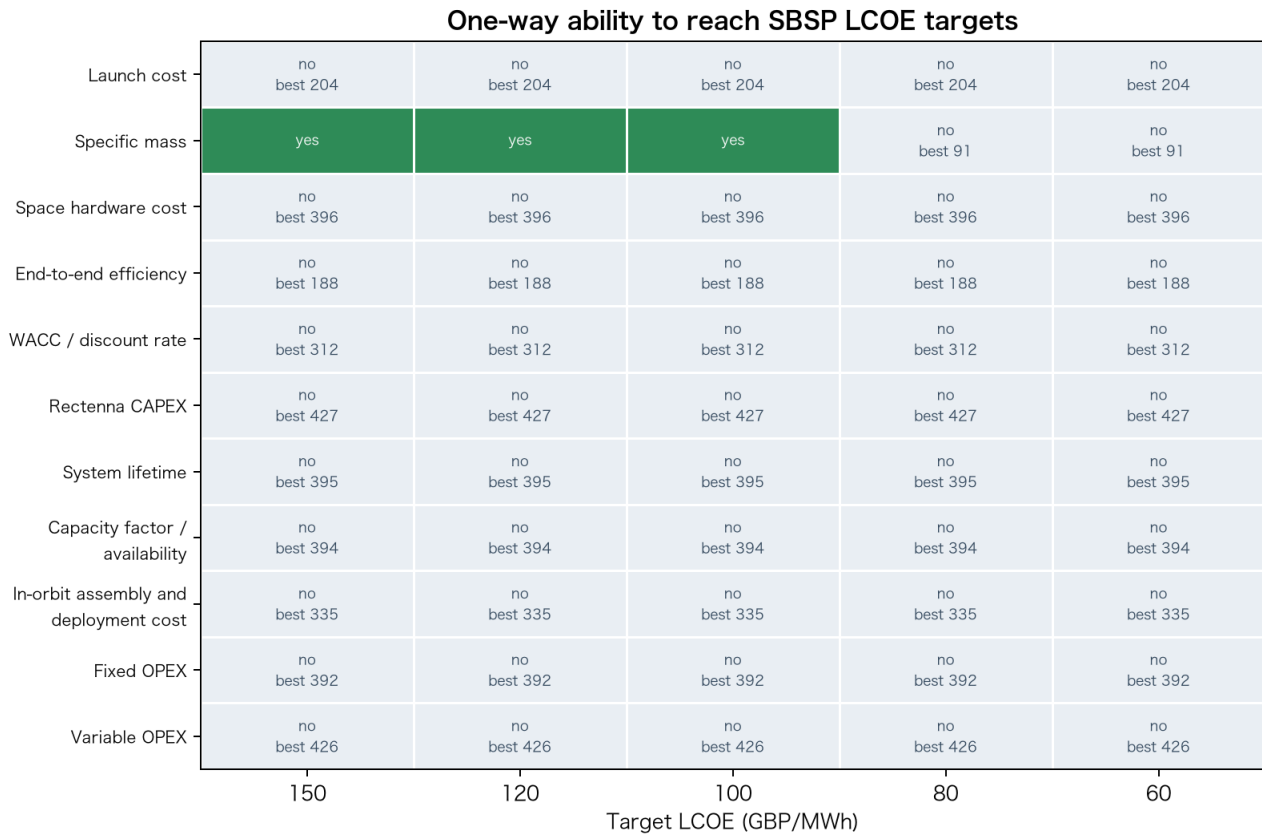
The main uncertainties are architecture feasibility, cost-range credibility, financing risk, lifetime and degradation, beam and regulatory constraints, and the value of firm output in a future GB system. Price-basis differences between the 2018 BEIS and 2024 DESNZ comparators remain explicit rather than silently harmonised.

The central conclusion is therefore narrow: SBSP does not become decision-relevant through launch-cost reduction alone. It becomes worth deeper assessment only when a credible integrated architecture enters roughly 80-120 GBP/MWh delivered-grid LCOE.

Appendix A. Full One-way Diagnostic

The report focuses on decision-relevant views. Full sensitivity curves remain in figures/, and all curve points remain in data/processed/sensitivity_curves.csv.

Figure A1. Full one-way feasibility matrix



The matrix retains the complete one-way audit across all five targets and eleven varied inputs.

Appendix Table A1. Full one-way threshold summary

Driver	Unit	150 threshold	100 threshold	Best one-way LCOE
Launch cost	GBP/kg to staging orbit	not reached	not reached	204
Specific mass	kg/kW-space	1.283	0.616	91
Space hardware cost	GBP/W-space	not reached	not reached	396
End-to-end efficiency	fraction	not reached	not reached	188
WACC / discount rate	fraction	not reached	not reached	312
Rectenna CAPEX	GBP/W-delivered	not reached	not reached	427
System lifetime	years	not reached	not reached	395
Capacity factor / availability	fraction	not reached	not reached	394
In-orbit assembly and deployment cost	GBP/kg	not reached	not reached	335
Fixed OPEX	fraction/year	not reached	not reached	392
Variable OPEX	GBP/MWh	not reached	not reached	426

Appendix B. Source Registry

Appendix Table B1. Source registry

Source ID	Reference	URL
DESNZ_EGC_2025	DESNZ, Electricity generation costs 2025	https://www.gov.uk/government/publications/electricity-generation-costs-2025
BEIS_EGC_2020	BEIS, Electricity Generation Costs 2020	https://www.gov.uk/government/publications/beis-electricity-generation-costs-2020
NESO_BALANCING_2025	NESO, 2025 Annual Balancing Costs Report	https://www.neso.energy/document/362561/download
NESO_NETWORK_UPDATE_2026	NESO, Beyond 2030 Electricity Transmission Update	https://www.neso.energy/publications/beyond-2030
NESO_CP2030	NESO, Clean Power 2030 advice and implementation material	https://www.neso.energy/publications/clean-power-2030
NESO_OPERABILITY_2026	NESO, Operability Strategy Report and Electricity Markets Roadmap	https://www.neso.energy/publications/operability-strategy-report-and-electricity-markets-roadmap
UKSA_S BSP_2020	UK Space Agency and BEIS, SBSP research commission	https://www.gov.uk/government/news/uk-government-commissions-space-solar-power-stations-research
CALTECH_S BSP_2022	Caltech SBSP technical concept paper 2	https://arxiv.org/abs/2206.08373
NAO_HPC_2017	National Audit Office, Hinkley Point C	https://www.nao.org.uk/reports/hinkley-point-c/

Appendix C. Assumption Classification

All reference inputs in Table C1 are classified as exploratory modelling assumptions. Operating and grid-interface allowances remain documented in data/sbsp_parameters.csv; raw official workbooks and processed CSVs remain separate from model outputs.

Appendix Table C1. Decision-relevant reference inputs

Parameter	Reference	Source	Interpretation
Delivered grid capacity	2000.000 MW	ASSUMPTION_THIS_STUDY	Scale anchor. LCOE is mostly scale-neutral because most costs are specified per W or per kg.
End-to-end efficiency	15.0%	CALTECH_S BSP_2022	Exploratory conversion from required space-side power to delivered grid output.
Capacity factor / availability	90.0%	ASSUMPTION_THIS_STUDY	Availability assumption for a near-firm output profile, not measured SBSP operating history.
Specific mass	5.000 kg/kW-space	CALTECH_S BSP_2022	Architecture-level mass intensity. This is the strongest one-way driver and remains highly uncertain.
Launch cost	500.000 GBP/kg to staging orbit	UKSA_S BSP_2020	All-in launch-cost threshold variable. Treated as a wide exploratory range rather than a forecast price.
Space hardware cost	0.400 GBP/W-space	ASSUMPTION_THIS_STUDY	Exploratory manufacturing-cost threshold for orbital collection, structure, control and conversion hardware.
WACC / discount rate	6.5%	DESNZ_EGC_2025	Financing-cost variable anchored to infrastructure and higher-risk project-finance bounds.
System lifetime	30.000 years	ASSUMPTION_THIS_STUDY	Economic operating life. Space degradation and refurbishment are represented separately.
In-orbit assembly and deployment cost	200.000 GBP/kg	UKSA_S BSP_2020	Exploratory robotic assembly, deployment, inspection and commissioning cost.
Rectenna CAPEX	0.150 GBP/W-delivered	ASSUMPTION_THIS_STUDY	Exploratory ground receiving and conversion cost per delivered W.